IV_SENTINEL BY GROUP 20

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Problem

High patient to nurse ratio
Life-threatening complications
Medication errors

Project Objectives

Direct Mass Sensing
Multi-Threshold Alerting
Transmit real-time medical data through thing speak to central dashboard

Project Requirements

1. ESP32 Dev Board
2. Load cell(5kg)
3. HX711
4. 12C 16*2 LCD
5. LEDs(Red, Blue, Green)
6. Buttons
7. Breadboard + jumper wires
8. 3D printed Enclosure
9. External 5V/2A supply
10. A demo stand to support setup

Target Users

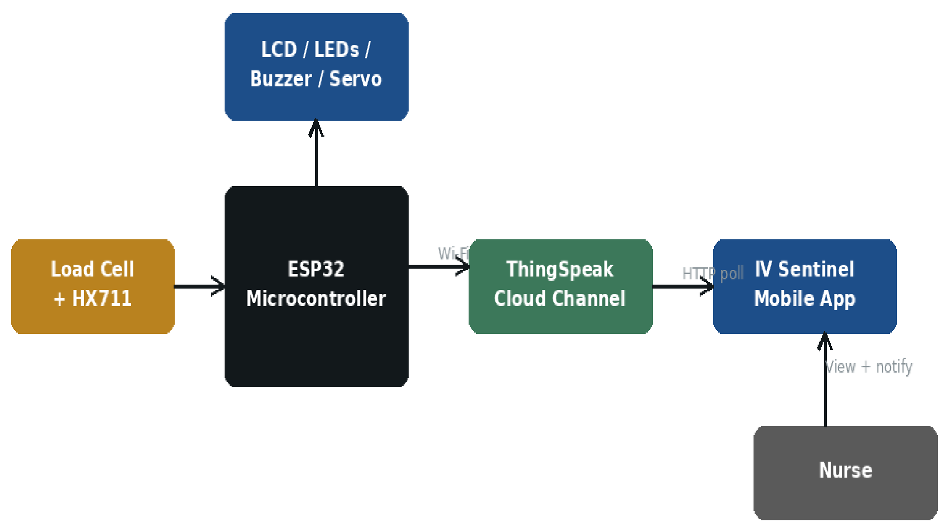
DOCTORS AND NURSES

- Helps staff to monitor assigned patient beds and minimizes the risks of air entering the IV tube while away.

PATIENTS

- Allows patients to monitor level of the liquid in the IV bag so that notify the medical staff in time if assistance is needed.

Project Design



Bedside unit (left) operates independently of Wi-Fi/cloud (right).

he bedside unit uses an ESP32 with a load cell and HX711 for continuous mass-based IV fluid monitoring. Real-time data is shown on an LCD, with RGB LEDs (Green/Orange/Red) indicating alert levels. The unit operates independently of Wi-Fi, while optional ThingSpeak cloud integration enables remote viewing via the IV Sentinel mobile app for staff notification.

Results

Step 1

- Initial calibration of load cell with HX711
- Zero-balancing and weight offset adjustment
- Stable baseline reading established
- System powered on and sensor responsive

Step 2

- IV bag at optimal fill level
- Fluid percentage stable and within safe range
- Green LED indicates normal condition
- Real-time mass data transmitted to dashboard

Step 3

- Fluid level approaching threshold
- Orange alert triggered – IV bag below 30%
- Warning notification sent to central dashboard
- Nurse alerted for timely replacement

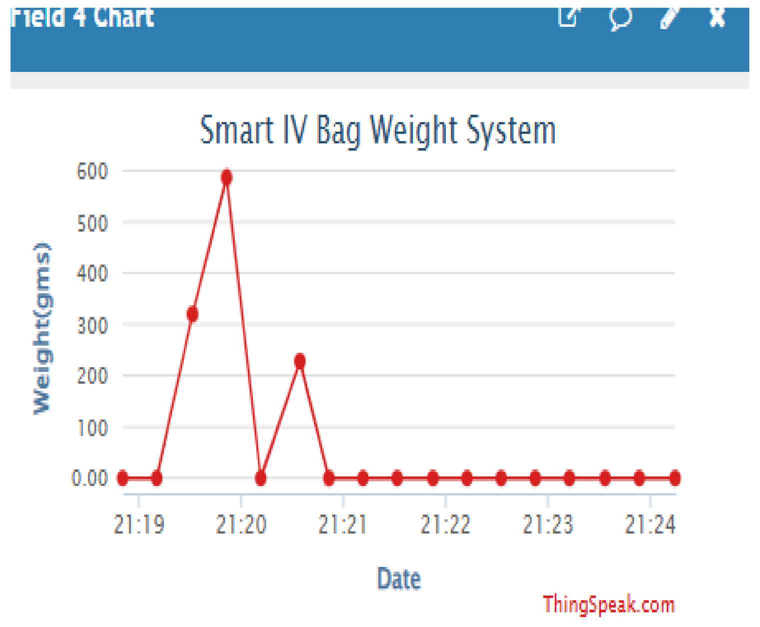
Step 4

- Critical fluid level detected – below 10%
- Red LED activated – emergency alert
- Immediate notification displayed on LCD
- System prompts urgent staff intervention

Future Work

1. Thermal Drift Compensation
2. Adaptive Multi-Bag Calibration
3. Hospital Nurse Call Integration

Results



The video above shows how the percentage of the fluid in the IV bag changes with time .

The color green shows that the fluid in the IV bag is at a good level, orange indicates that the fluid is low and lastly red represents a critical condition.

Conclusion

Experimental findings confirm the hypothesis that mass-based load cell sensing coupled with 3-point tare calibration overcomes the fundamental failures of optical drop counters and ultrasonic level sensors

References

- **[5] K. Rajesh, P. Anand, and R. Suresh**, "Automated IV drip rate monitoring using infrared sensing and GSM alerting," in *Proc. IEEE Int. Conf. on Communication and Signal Processing (ICCSP)*, Chennai, India, Apr. 2017, pp. 1423–1427.
- **[7] R. Patel, V. Mehta, and S. Joshi**, "Deep learning-based visual classification of intravenous drip levels using ESP32-CAM," *IEEE Access*, vol. 10, pp. 45231–45242, 2022.
- **[14] P. Rao and S. Venkat**, "Wi-Fi enabled load cell system for remote IV bag weight monitoring in ICU settings," *Journal of Medical Devices*, vol. 15, no. 2, pp. 021003–021009, Jun. 2021.
- **[17] B. Naidu and R. Krishnan**, "Multi-bed IV drip monitoring with AWS IoT Core and Raspberry Pi for smart hospital infrastructure," *IEEE Internet of Things Journal*, vol. 9, no. 11, pp. 8912–8924, Jun. 2022.